

# The $d = 2$ equal-exponent theorem with a nonconstant amplitude

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2026-09-07

## Abstract

For the two-dimensional chart integral with general amplitude  $\Phi(u, v, s)$  and equal exponents  $(h_1 + 1)/k_1 = (h_2 + 1)/k_2 = p$ , the anchored decomposition  $\Phi = \Phi(u, 0) + \Phi(0, v) - \Phi(0, 0) + [\Phi(u, v) - \Phi(u, 0) - \Phi(0, v) + \Phi(0, 0)]$  splits  $Z_\Phi$  into two axis integrals (unit 87), a corner integral, and a mixed term dominated by a constant-kernel block with both exponents shifted (unit 47, one logarithm lost). Hence

$$Z_\Phi(N) = N^{-p}(A \log N + B) + O(N^{-(p+\delta)}(1 + \log N)), \quad \delta = \min(1/k_1, 1/k_2),$$

with  $A = (k_1 k_2)^{-1} \int s^{p-1} e^{-\beta s^2} \Phi(0, 0, s) ds$  and  $B = (k_1 k_2)^{-1} \int s^{p-1} (\log R - \log s) e^{-\beta s^2} \Phi(0, 0, s) ds + k_2^{-1} \int s^{p-1} e^{-\beta s^2} I_1(s) ds + k_1^{-1} \int s^{p-1} e^{-\beta s^2} I_2(s) ds$ . This is the milestone of the second Astra consult: the first complete nonconstant- $\xi, \eta$  log polynomial of `thm:TaylorTree`, whose constant term sees both axes and not only the corner. Zero sorry/axiom.

## 1 The things formalised

```
theorem twoDamp_anchored ... :
  twoDamp b N h h k k  $\Phi$ 
    = twoDamp b N h h k k (fun u _ s =>  $\Phi$  u 0 s)
      + twoDamp b N h h k k (fun _ v s =>  $\Phi$  0 v s)
      - twoDamp b N h h k k (fun _ _ s =>  $\Phi$  0 0 s)
      + twoDamp b N h h k k (mixedAmp  $\Phi$ )
theorem twoDamp_mixed_le ... :
  |twoDamp b N h h k k (mixedAmp  $\Phi$ )|
  C * max 2 (4 / ) * twoDGeneral ( / 2) (2 * L) b (N ^ 2) (h + 1) (h + 1) k k
theorem twoDamp_second_order ( b L C C M p : ) (h h k k : ) ( $\Phi$  :  $\rightarrow \rightarrow \rightarrow$  )
  (h : 0 < ) (hb : 0 < b) (hk : 0 < k) (hk : 0 < k)
  (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p)
  (hC : 0 C) (hC : 0 C)
  (h $\Phi$  : Continuous fun x :  $\times \times$  =>  $\Phi$  x.1 x.2.1 x.2.2)
  (hlipA : u Icc (0 : ) b, s, 0 s  $\rightarrow$ 
    | $\Phi$  u 0 s -  $\Phi$  0 0 s| C * u * ((1 + s) * Real.exp ( * s * L)))
  (hlipB : v Icc (0 : ) b, s, 0 s  $\rightarrow$ 
    | $\Phi$  0 v s -  $\Phi$  0 0 s| C * v * ((1 + s) * Real.exp ( * s * L)))
  (hmix : u Icc (0 : ) b, v Icc (0 : ) b, s, 0 s  $\rightarrow$ 
    |mixedAmp  $\Phi$  u v s| C * (u * v) * ((1 + s) ^ 2 * Real.exp ( * s * L)))
  (h $\Phi$  : s, 0 s  $\rightarrow$  | $\Phi$  0 0 s| M * Real.exp ( * s * L)) :
  (fun N : => twoDamp b N h h k k  $\Phi$ 
    - N ^ (-p) * (twoDA p k k  $\Phi$  * Real.log N + twoDB b p k k  $\Phi$ )) =0[atTop]
  fun N : => N ^ (- (p + min (k : ) ^ 1 (k : ) ^ 1)) * (1 + Real.log N)
```

## 2 Proof strategy

All linearity is done in product form on  $(0, b]^2$  (`integral_prod`), where every piece is continuous on the compact box. The two axis pieces and the corner piece are instances of the axis expansion of unit 87 (the second after the coordinate swap of unit 86), giving  $N \geq 1$  bounds with exponents  $p + 1/k_1$ ,  $p + 1/k_2$ , converted to  $O(N^{-(p+\delta)}(1 + \log N))$ . The mixed piece is bounded pointwise by  $C_2 uv(1 + s)^2 e^{\beta L s}$  and  $(1 + s)^2 \leq \max(2, 4/\beta) e^{(\beta/2)s^2}$ , hence by the constant-kernel integral with exponents  $(h_1 + 1, h_2 + 1)$ , whose asymptotics `twoDGeneral_isBigO_min` at  $n = N^2$  read  $O(N^{-(p+\delta)} \log N)$ . The coefficients add up because all three one-variable amplitudes agree at the corner, and the corner piece has vanishing axis integral.

## 3 Roadblocks and resolutions

- `Chained`  $\leftarrow$  `integral_add`/ $\leftarrow$  `integral_sub` rewrites fail on the `Pi.add` form of `Integrable.add`; give the combined integrability facts single-lambda type ascriptions.
- `twoDIntegrand_integrable` cannot infer the amplitude from a continuity proof of its composite; pass the amplitudes explicitly.
- `positivity` needs  $0 < u, 0 < v$  as hypotheses, not membership in `Ioc`.

## 4 Outcome

Astra's step 5 milestone is complete at the amplitude level. Remaining polish: derive the amplitude hypotheses from  $C^2$  regularity of  $\xi, \eta$  for  $\Phi = \eta e^{\beta s \xi}$  (mixed second difference), the  $n$ -form  $Z(n) = n^{-p/2}(\frac{A}{2} \log n + B) + O(n^{-(p+\delta)/2}(1 + \log n))$ , and the face-parameter second-order theorem under the strict gap.